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Table of Contents

Mevhibe Ay Türkmen & Nurgül Evcim*

An Analysis of the Logistic Performances of Countries

Abstract In the global world, the creation of the supply chain has made it possible to use reliable, cheap, and fast transportation. It has also become more difficult to get into competition. In this environment, businesses are supposed to make economic and qualified production as well as provide a more effective service. They are also obliged to deliver the products whenever and wherever the clients demands and also meet their expectations in terms of quality. Thus, logistic activities have become a measurement of performance that directly influences the competitive capacities of businesses. For these reasons, logistics play a major role in the world economy and it is one of the most important tools of world trade. It is even more important for developing countries, which have a big potential for growth. This study includes 27 countries that have higher rates in the Logistics Performance Index (LPI) and are described as 'high income countries' according to the LPI rating. The study will measure the effect of the macroeconomic indicators of these countries on the LPI using regression analysis. According to the results of the analysis, countries with higher LPI scores have better economies.

Keywords: Logistics, Logistics Performance Index, Economic indicators

Introduction

Logistics is a significant activity in today's sophisticated global economy. The market problem created by continuous manufacturing, which was initiated with the industrial revolution, resulted in a competitive environment. In this period, markets were the most important factor that determined the production, overshadowing customer expectations. As a result of globalisation in the post-1980 period, the development of technology enabled businesses to access international markets. The formation of the global supply chain accompanied reliable, inexpensive, and rapid transportation opportunities. Thus, every corporation in logistics

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needs to conduct its business faster and in a more reliable manner to stay alive (Özmen, 2008: 2). Furthermore, it became even harder to be competitive. In a study sponsored by the World Bank and conducted on the measurement of logistics performances and costs of certain research organisations, Rantasila and Ojala (2012) reported that this performance and cost assessments were important for national and regional policy making to gain competitive advantages. The requirements for the businesses to conduct economic and quality manufacturing, to provide efficient service quality, and to deliver their goods wherever and whenever the customers desire with quality delivery services resulted in logistics activities to become a significant performance criterion that affects the competitiveness of the businesses directly. Distance is no longer a factor that increases the cost of international economic relationships and information exchange and economic activities became minute affairs (Oda, 2008: 23). In a study by Babacan (2003) that scrutinised the current status of the Turkish logistics industry, interviews with executive managers in the industry demonstrated that the significance of the industry increased especially during recent years, and it was necessary to reduce the costs and enlarge the market.

Logistics was defined in different manners, since it has a wide area of application. With the effect of globalisation and the development of technology, definitions of logistics differed as well. The etymology of logistics goes back to the Latin "logic" and "static" (statistics), which when combined means "logical statistics" (Koptekin, 2002: 46). The most accurate definition of logistics was made by the Council of Logistics Management: "the process of planning, implementing, and controlling the efficient, effective flow and storage of goods, services, and related information from point of origin to point of consumption for the purpose of conforming to customer requirements." In its broadest sense, logistics is the total of all activities conducted to deliver the goods from the source to the final consumer. These activities are conducted within a flow called the supply chain.

Logistics is also one of the important competitive advantages in the foreign trade of countries. Commercial activities have spread all around the world as a result of globalisation. Thus, logistics became global as well, since trade could not exist without logistics (Şahin, 2014: 348). Foreign trade logistics has an additional significance for developing countries to control global trade and to get a share of the benefits of globalisation (Kara et al., 2009: 71). Thus, there is a cyclical relationship between the two. Ateş and Işık (2010) analysed the effects of the developments in logistics services in Turkey on exports, and determined a bi-directional causality relationship between the logistics industry and exports using the Granger causality analysis. As the international trade increases, logistics activities increase as well,

thus developments occur in the logistics industry. These developments in the logistics industry in turn facilitate international trade, increasing commercial activities. Thus, for national trade to reach a good performance within the transportation system, logistics became one of the most significant variables. The World Bank publishes an important indicator that reflects the logistic performances of countries at the international level. The Logistic Performance Index (LPI) is a comprehensive index aimed to assist the nations to determine difficulties and opportunities they faced in their international trade performances. This study would scrutinise the fundamental factors that affect or constitute logistics performance.

1. What is the Logistic Performance Index?

In the global economy, the development of trade is dependent on the development of the logistics industry. As the logistics industry develops, trade would develop and as trade develops, logistics services would develop further as a result. This is the reason why the World Bank specified the logistic performance for each country. The information obtained from the web-based research findings that is collected from several international companies in the logistics industry form the framework of the Logistic Performance Index (LPI) published initially in 2007 and republished every two years since then. The index is created using the points collected in the survey that company workers and managers participate in. The measurement is conducted in two forms based on global and domestic criteria. The following six criteria are considered for the global measurement (World Bank, 2007):

- Customs clearance (the efficiency of customs and border management clearance).
- Infrastructure (seaports, railroads, information technology development).
- International transportation.
- The quality of logistics services.
- Domestic logistic costs (or tracking and tracing).
- Timely delivery (the frequency with which shipments reach consignees within scheduled or expected delivery times).

Study data were collected from more than 800 of the best logistics companies in the world (World Bank, 2007). The scoring used in the measurement changes between the values of 1 and 5. 1 depicts the lowest possible score, where 5 is the highest possible score. Low scores in at least two of the six criteria above would result in lower logistics performance scores. Generally under-developed countries share lower tiers, while the developed countries find themselves on the top of the list. The development of logistics depends on the logistic potential and abilities of countries. Country and region-based logistic evaluations are important, since certain parts of the world have excellent logistic opportunities, which contribute to their success extremely, while others lack these characteristics (Tutar et al., 2009: 196). Thus, it could be argued that these performance indicators are significant for the comparison and interpretation of economic indicators of countries.

Table 1 contains LPI scores of "logistics friendly" (Top Quintile) countries published in 2014 by the World Bank. The World Bank defined the countries that received a score of 3.51 or higher out of 5 as top logistic performance quintile. In the study conducted by measuring the logistics activities in the participating 160 countries, the scores of the top ten countries did not demonstrate significant changes within the years. Although their order might change, the countries that are in this tier are always the same. Singapore, which was first between 2007–2012 was replaced by Germany in 2010–2014. Germany is followed by high-income countries such as the Netherlands, Belgium, and the United Kingdom. An analysis of the fundamental criteria of the study demonstrated that Norway was first in customs value and logistics efficiency and Germany was leading in infrastructure and timeliness criteria. Luxembourg led in international transportation and timely delivery criteria. It could be argued that the advantages such as the development level of countries, advanced level of their logistics industries, low transportation costs due to geographical advantages, low delivery times are the main factors to get better scores in the logistic performance index. The fact that leading countries in the reports, which have been published since 2007, were almost always the same signifies the importance of logistics in development. Thus, Çevik and Gülcan (2011) declared the logistics industry the engine of national development.

Table 1: International LPI Values of High-Income Countries

Countries	LPI		Customs		Infrastructure		International Shipments		Logistics Quality and Competence		Tracking and Tracing		Timeliness	
	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank
Germany	4.12	1	4.10	2	4.32	1	3.74	4	4.12	3	4.17	1	4.36	4
Netherlands	4.05	2	3.96	4	4.23	3	3.64	11	4.13	2	4.07	6	4.34	6
Belgium	4.04	3	3.80	11	4.10	8	3.80	2	4.11	4	4.11	4	4.39	2
U.Kingdom	4.01	4	3.94	5	4.16	6	3.63	12	4.03	5	4.08	5	4.33	7
Singapore	4.00	5	4.01	3	4.28	2	3.70	6	3.97	8	3.90	11	4.25	9
Sweden	3.96	6	3.75	15	4.09	9	3.76	3	3.98	6	3.98	7	4.26	8
Norway	3.96	7	4.21	1	4.19	4	3.42	30	4.19	1	3.50	31	4.36	5
Luxembourg	3.95	8	3.82	10	3.91	15	3.82	1	3.78	14	3.68	22	4.71	1
U. States	3.92	9	3.73	16	4.18	5	3.45	26	3.97	7	4.14	2	4.14	14
Japan	3.91	10	3.78	14	4.16	7	3.52	19	3.93	11	3.95	9	4.24	10
Ireland	3.87	11	3.80	12	3.84	16	3.44	27	3.94	9	4.13	3	4.13	16
Canada	3.86	12	3.61	20	4.05	10	3.46	23	3.94	10	3.97	8	4.18	11
France	3.85	13	3.65	18	3.98	13	3.68	7	3.75	15	3.89	12	4.17	13
Switzerland	3.84	14	3.92	7	4.04	11	3.58	15	3.75	16	3.79	18	4.06	21
Hong Kong	3.83	15	3.72	17	3.97	14	3.58	14	3.81	13	3.87	13	4.06	18
Australia	3.81	16	3.85	9	4.00	12	3.52	18	3.75	17	3.81	16	4.00	26
Denmark	3.78	17	3.79	13	3.82	17	3.65	9	3.74	18	3.36	36	4.39	3
Spain	3.72	18	3.63	19	3.77	20	3.51	21	3.83	12	3.54	26	4.07	17
Taiwan	3.72	19	3.55	21	3.64	24	3.71	5	3.60	25	3.79	17	4.02	25
Italy	3.69	20	3.36	29	3.78	19	3.54	17	3.62	23	3.84	14	4.05	22
Korea Rep.	3.67	21	3.47	24	3.79	18	3.44	28	3.66	21	3.69	21	4.00	28
Austria	3.65	22	3.53	23	3.64	25	3.26	40	3.56	26	3.93	10	4.04	23
New Zealand	3.64	23	3.92	6	3.67	22	3.67	8	3.56	27	3.33	38	3.72	40
Finland	3.62	24	3.89	8	3.52	28	3.52	20	3.72	19	3.31	39	3.80	38
Malaysia	3.59	25	3.37	27	3.56	26	3.64	10	3.47	32	3.58	23	3.92	31
Portugal	3.56	26	3.26	31	3.37	31	3.43	29	3.71	20	3.71	20	3.87	35
United Arab Emirates	3.54	27	3.42	25	3.70	21	3.20	43	3.50	31	3.57	24	3.92	32
China	3.53	28	3.21	38	3.67	23	3.50	22	3.46	35	3.50	29	3.87	36
Qatar	3.52	29	3.21	37	3.44	29	3.55	16	3.55	28	3.47	32	3.87	34
Turkey	3.50	30	3.23	34	3.53	27	3.18	48	3.64	22	3.77	19	3.68	41

Kaynak: World Bank, (2014) http://lpi.worldbank.org/international/global

The table data are the results of the following questions posed (World Bank, 2007):

1. What is the efficiency of customs and border management clearance operations including the customs administration?
2. What is the level of infrastructure related to international transportation?
3. What is level of the ease of shipments in international transportation?
4. Please evaluate the competence and quality of logistics services.
5. What is the level of the ability to track and trace consignments?
6. What is the frequency of timely delivery within the scheduled time?

These criteria were evaluated within the 1–5 points range among each other. Norway was the first in the efficiency of customs and border management clearance operations with 4.21 points. It was followed by Germany and Singapore. Turkey was 34th with 3.23 points in the range of upper-medium tier. In the infrastructure level, Germany was first with 4.32 points, Singapore was second with 4.28 points, and Netherlands was third with 4.23 points. Turkey was 27th with a not so bad score of 3.53. The country with the highest ease of consignments was Luxembourg. Luxembourg got 3.82 points and was followed by Belgium with 3.80 points. Turkey was 48th with 3.18 points. The countries with the highest competence and quality of logistics services points were Norway with 4.19 points, again Netherlands with 4.13 points, and Germany with 4.12 points. Turkey was ahead of Italy, Austria, and China in 22nd place with 3.64 points. The countries with high scores in the ability to track and trace consignments were Germany with 4.17 points, the USA with 4.21 points, and Ireland with 4.13 points. Turkey was 19th with 3.77 points ahead of Norway, Luxembourg, Denmark, and Spain. Finally, the countries with the highest frequency of timely delivery within the scheduled time were Luxembourg with 4.71 points, Belgium with 4.39 points, and Denmark, which was behind in other criteria, with 4.39 points. Turkey was 41st with 3.68 points.

Turkey was 34th with 3.15 points in the 2007 LPI report, in the 2010 report, it fell behind to 39th place with 3.22 points, and rose to 27th place with 3.51 points in the 2012 report. Finally, Turkey was behind again in 30th place with 3.50 points in 2014. There were certain developments in the logistics industry between 2007 and 2014. Turkey was hopeful in the sense that it occupied the last place among high-income countries, ahead of the European Union member states such as Poland, Bulgaria, Hungary, Cyprus, Czech Republic, and Romania in 2012 and 2014.

2. The Factors That Affect Logistic Performance

The logistics industry, which increasingly becomes more significant, is also important for national economies. It is one of the most important factors for the development of foreign trade. Hausman, Lee, and Subramanian (2013) measured the effects of logistics as the countries conduct their import and export activities in global trade on the costs, time, and reliability. They have concluded that, as the logistic performance improves, the trade would increase as well. However, as mentioned before, there is a cyclical relationship between foreign trade and logistics. Thus, as the foreign trade improves, logistic activities would also develop.

Among the fundamental factors that affect the logistic activities of countries, the number of countries that participated in the survey study by the World Bank could be considered as the most significant. A high number of participants would help increase the possibility of obtaining accurate information. In addition, national economic indicators are also among the significant factors. In other words, high values in developmental parameters would also positively affect the logistic activities.

The leading countries in the LPI reports all have high GDP ratios as well. Analysis results based on EU and USA data show that the share of logistics in GNP is about 10% (Bayraktutan et al., 2012: 62). Thus, a review of the table would demonstrate that, while countries with high national income are in the "logistics friendly" group, countries with low income are placed in the low tier groups with poor logistic activities. However, the differences in logistic performance between the countries with similar income level demonstrate that income is insufficient in explaining the performance alone. It was considered that the total income of non-resident citizens of the countries could be effective on the logistic performance. In addition to income, the national trade volume, the share of imports and exports in domestic income, industrial added value, and services industry added value affect the performance the most. To measure the effects of these factors on the LPI, a simple regression analysis would be conducted. As a result of this analysis, an attempt would be made to measure whether the above mentioned economic indicators "were effective in LPI, and if positive in which extent."

3. Econometric Method and Modelling

The objective of the study to answer the questions; "is there are relationship between LPI and economic indicators? And if there is, what is the strength of this relationship?" For this purpose, the 2011 macroeconomic indicators were used for comparison with the 2012 LPI data in the econometric analysis. Cross-section data were utilised in the study. The 27 countries considered as logistics friendly (high performance) by the World Bank were included in the model. The LPI was considered as the dependent variable, while explanatory variables were the share of imports and exports in GDP (%), the share of the trade in goods in GDP (%), the ratio of industrial added value and services industry added value in GDP (%), annual GDP growth, and GNP ratio (Gogoneata, 2008: 145). All data were procured from the World Bank database.

The econometric method was based on the single variable linear regression model. A different regression model was used for each variable. The OLS (Ordinary Least Square) was used as an estimator. All data were run in the E-Views software, and the results (coefficients, t-statistics, and standard error) were interpreted within a 5% significance level. The equation was as follows (Gogoneata, 2008: 145):

$$LPI = \alpha + \beta X + \varepsilon$$

LPI: Logistic Performance Index
X: GDP, GNI, EX(% GDP), IM(%GDP), TRA(%GDP), IDS(%GDP), SER(%GDP)

Table 2: Explanation of Independent Variables

Independent Variables	Explanation
GDP	Gross Domestic Product Growth (annual %)
GNI	Gross National Income (with Atlas Metodu)
EX	Exports of Goods and Services (% of GDP)
IM	Imports of Goods and Services (% of GDP)
TRA	Merchandise Trade (% of GDP)
IND	Industry, value added (% of GDP)
SER	Services, value added (% of GDP)

Table 3: Relationship Between LPI and Economic Indicators

Independent Variables	Dependent Variable = LPI					
	A		β		R^2	N
GDP	3.8917 *	(69.8109) [0.000]	-0.0203	(-1.3526) [0.1888]	0.0708	26
GNI	3.8680 *	(78.3884) [0.000]	-0.0036	(-0.2008) [0.8430]	0.0022	20
EX	3.7527 *	(75.6050) [0.000]	0.0012 *	(2.1477) [0.0420]	0.1612	26
IM	3.7430 *	(75.2206) [0.000]	0.0016 *	(2.3690) [0.0262]	0.1895	26
TRA	3.7327 *	(80.9574) [0.000]	0.0011 *	(2.9152) [0.0076]	0.2615	26
IND	4.0441 *	(41.3831) [0.000]	-0.0076 *	(-2.3118) [0.0301]	0.1885	25
SER	3.2424 *	(16.1233) [0.000]	0.0084 *	(2.9689) [0.0069]	0.2770	25

*5% significant, (t statistics), [p value]

While the explanatory coefficients obtained as a result of the regression analysis provide information about the explanatory power of the independent variables for the dependent variables, and the other coefficients (α, β) provide information on the effects of possible changes in the variables on the dependent variable. The explanatory variables obtained in the study varied between 0.2% and 27.70%.

The most effective variable in explaining the logistic performances of the countries included in the study was the SER (Services Industry Added Value share in GDP), which alone explained 27.70% of the LPI. At the same time, when the SER increases by 1 unit, there will be an increase of 0.0084 in the LPI, since it is a high elasticity variable. Thus, it could be argued that the national logistic performance and services sector added value have a positive relationship. Therefore, a strong and developed services industry in developed and developing countries would result in the development of infrastructure services and efficient operation of the institutions in the country. Thus, the LPI would be affected positively by the changes in the services industry. Another variable that has a strong effect in explaining the logistic performance was the national trade of goods volume. It explained 26.15% of the logistic performance. A 1-unit increase in the trade of goods would cause an increase of 0.001 in the logistic performance. Thus, an increase in the trade of goods of the countries would improve logistics due to the effects related to foreign trade. Import and export shares both have a high explanatory power on the logistic performance. While imports explained 18.95%, exports explained 16.12% of the logistic performance. As mentioned in the section about trade of goods, as imports and exports increase, logistic activities and performance of the countries in the international arena would increase as well. As a result, their infrastructure would improve, and cyclically the imports and exports would increase. Industrial added value is one of the variables that affect the logistic performance with an explanatory power of 18.85%. However, there is a negative relationship between the industrial added value and logistic performance. When industrial added value increases 1 unit, the logistic performance would decrease by 0.0076. This was an unexpected development. Normally, industrialisation of nations would develop their foreign trade, which would develop their logistic activities in return.

The GDP growth had a relatively lower effect when compared to other variables with an explanatory power of 7.08%. Furthermore, due to the negative value of gradient coefficient, when the national GRP growth rate increases, the logistic performance would move to the opposite direction, and the LPI would decrease. Despite the contrary expectations, the occurrence of such a result increases the significance of its reasons. Countries might experience fast growth rates every year, or developed and developing countries could face lower growth rates. When explaining growth, it is possible to mention various indicators such as direct foreign investments, bank loans, savings, foreign trade rate, etc. However, even though these indicators immediately affect growth, they would not immediately affect infrastructure and institutions, which are significant for national logistic activities, since countries prefer natural resources and cheap labour in manufacturing. On the other hand, developed countries do not have infrastructure problems in the first place. Their logistic performances would move with less dependency whether their growth is slow or fast. Finally, the GNP ratios of the countries affect the LPI negatively. With an explanatory power of 0.2%, a 1 unit increase in GNP would cause a 0.0036 decrease in logistic performance. As a result, there is a negative relationship between logistic performance and growth, the GNP and industrial added value. When one of these indicators increases, there would be a decrease in the LPI.

Result

The logistics industry is one of the significant high-growth industries in the world. In addition to its effect on the cost factor of the economy with the added value it creates, it increases the competitiveness of the businesses and provides the advantage of sustainability. The study aimed to measure the effects of the national macroeconomic indicators on the logistic performances of 27 countries, which were considered as high-income countries based on the LPI results published by the World Bank. Among the variables scrutinised in the study and assumed to be effective on the LPI, the services industry added value rate and trade of goods variables had the most significant effects on logistic performance. Any increase in these variables would affect logistic performance positively. However, the GDP growth rate, GNP rate, and industrial added value rate variables produced findings against the expectations. While it was considered that these variables would normally affect logistic performance positively previously, it was found that these

variables had negative effects. It is possible to interpret these results concluded only through the findings of this study as follows:

The countries included in the study were either developed countries that guide the world economy, or developing countries. These were countries with seriously high economic indicators across the board, from GDP to imports and exports, from industrial added value to services sector added value. The finding that growth had a negative relationship with the LPI is not sufficient to conclude that there is an actual negative relationship between these two. However, although study results would show that growth had no direct effect on the LPI, it would be impossible to ignore its indirect effects. A nation could have grown quite rapidly. This could have been due to various reasons. For instance, it could have been receiving increasing foreign investments, foreign trade, or bank loans could have been increased, or even production could have boomed. When this is the case, a sudden change in growth would not directly affect logistic performance. As a result of this growth, production could increase, imports and exports, hence the foreign trade could increase, where it is known that foreign trade and logistics industry affect each other cyclically. Thus, the logistic performance, which was not affected by growth directly, would be positively affected indirectly via the foreign trade channel. However, not only the foreign trade channel plays a role in this relationship, since it was possible to observe which countries were weak in which areas, and in which areas they should innovate, in the published LPI data. Thus, growth, albeit indirect, would contribute to the development of logistic performance with the investments in areas that are in need. Nuzzi (2010) stated that the increasing share of the private businesses in infrastructure development financing would be the awaited solution for the public funding problems. Thus, nations could develop logistic activities by increasing their investments in infrastructure, which is the most significant of the six factors that affect logistic performance.

Finally, all economic indicators included in the study have a high effect on national LPIs. Thus, the higher the economic indicators in a country, the higher the LPI score would be for that country. However, in the list published by the World Bank containing the LPI scores, there are countries with the same income level, but with different LPI scores. It could be argued that this was due to geographical advantages or disadvantages a country might have. The lack of maritime transportation means for certain countries, or extended delivery times due to long distances, and resulting high costs, climate conditions, natural disasters, restrictive policies by other nations, customs tariff, etc. would affect the logistic activities of the countries adversely. Even the domestic logistic indicators such as prolonged transactions, high number of required documentation, excessive red tape would create negative effects on the performance. Experts argue that, when the logistics institutions are privatised as opposed to run by public institutions, problems such as delays or high costs would disappear considerably. According to the data collected and published by professionals biannually could assist governments in the determination of loopholes and required rectifications in the logistics industry. They could develop appropriate policies and implement innovations in certain areas based on these data. Simply put, they could prioritise infrastructure investments, which is the most significant performance criterion in the international arena.

References

Ateş, İ. & Işık, E. (2010). "The Effects of Development in Logistics Services on Export Led Growth in Turkey", *Journal of Economic Sciences*, 2(1).

Babacan, M. (2005). "Development and Competitiveness Vision of Logistics in Turkey", *Izmir Vocational School Marketing Program.*

Bayraktutan, Y., Tüylüoğlu, Ş. & Özbilgin, M. (2012)."Analysis in Sector and Logistic Development Index: The Case of Kocaeli", *International Journal of Alanya Faculty of Business*, 4(3), 61–71.

Chow, G, Heaver, T.D. & Henriksson, L.E. (1994). "Logistics Performance: Definition and Measurement", *International Journal of Physical Distribution & Logistics Management*, 24(1), 17–28.

Cevik, O. & Gülcan, B. (2011). "Environmental Sustainability of Logistic Operations and Marco Polo Program", *Karamanoğlu Mehmetbey University Journal of Faculty of Economics and Administrative Sciences*, 20(1).

Gogoneata, B. (2008)."An Analysis of Explanatory Factors of Logistics Performance of a Country", *Journal of EconomicLiterature.*

Hausman, W.H., Lee, H.L. & Subramanian, U. (2005). "Global Logistics Indicators, Supply Chain Metrics, and Bilateral Trade Patterns", *World Bank Policy Research* Working Paper No:3773.

Hausman, W.H., Lee, H.L. & Subramanian, U. (2013). "The Impact of Logistics Performance on Trade", *Production and Operations Management*, 22(2), 236–252.

Kara, M., Tayfur, L. & Basık, H. (2009)."The Importance of Logistic Centers in Global Trade and Turkey", *Mustafa Kemal University Journal of the Institute of Social Sciences*, 6(11), 69–84.

Koptekin, M. (2002). "The Definition and Importance of Logistics", *Logistical Journal*, 1.

Nuzzi, A. (2010). "The Performance of the National Logistic Systems: What Strategies to Reduce the ItalianGap?", *Munich Personal RePEc Archive*, Paper No: 33374.

Oda, S. (2008). "Logistics Sector in Turkey and Its Effects on Foreign Trade", Master's Thesis, *Trakya University Institute of Social Sciences.*

Özmen, R.E. (2008). "World Economic Crisis and Logistics Sector", *Beykoz Logistics Vocational School.*

Persson, G. & Virum, H. (2001). "GrowthStrategiesforLogistics Service Providers: A Case Study", *The International Journal of Logistics Management*, 12(1), 53–64.

Portugal-Perez, A. & Wilson, J. S. (2012). "Export Performance and Trade Facilitation Reform: Hard and Soft Infrastructure", *World Development*, 40(7), 1295–1307.

Rantasila, K. & Ojala, L. (2012). "Measurement of National Level Logistic Cost and Performance", *International Transport Forum.*

Sahin, V. (2014). "An Evaluation on Logistic Geography", *Marmara Geography Journal*, 29, 344–362.

Töyli, J., Hakkinen, L. & Naula, T. (2008). "Logistics and financia performance: An analysis of 424 Finnish small and medium-sized enterprises", *International Journal of Physical Distribution & Logistics Management*, 38(1), 57–80.

Tutar, E., Tutar, F. & Yetişen, H. (2009). "A Comparative Analysis of Turkey's Logistic Sector to EU Member Countries (Romenia and Hungary)", *Karamanoğlu Mehmetbey University Journal of Faculty of EcDonomics and Administrative Sciences*, 17(2).

World Bank (2007). The Logistic Performance Index and It's Indicators.

World Bank (2010). The Logistic Performance Index and It's Indicators.

World Bank (2012). The Logistic Performance Index and It's Indicators.

World Bank (2014). The Logistic Performance Index and It's Indicators.

Yıldız, M.S., Bilgin, Y. & Yazgan, H.İ. (2013). "Analysis of the Factors Leading Firms to Invest in Logistic Activities: Example of Çınar Boru Profil Industry and Trade Inc.", *Business and Economics Research Journal*, 4(4), 131–145.